

# Flatcar Diagnostics

Source section: `diagnostics/` (renamed from `setup/debug/` in the refactored branch).

## Installing Debugging Tools

The Flatcar base image deliberately omits general-purpose debugging utilities to keep the footprint minimal. Use `toolbox` to access them:

```
toolbox
```

`toolbox` drops you into a Fedora container with the host filesystem accessible at `/media/root`. You get `strace`, `gdb`, `tcpdump`, package management, and other tools without modifying the base OS.

Inside toolbox:

```
# Access host filesystem
ls /media/root/etc
# Run a host binary
chroot /media/root /usr/bin/some-tool
```

## Reading the System Log

```
# All logs from current boot
journalctl -b

# Logs for a specific unit
journalctl -u docker.service

# Follow live
journalctl -f

# Kernel messages only
journalctl -k
```

Persistent logs are stored in `/var/log/journal/` if the directory exists (created by default on Flatcar).

## Collecting Crash Logs

Core dumps and crash logs go to `/var/log/crash/` . Configure `systemd-coredump` for structured core dump capture:

```
# Check for recent coredumps
coredumpctl list
coredumpctl info PID
```

## btrfs Troubleshooting

Flatcar uses btrfs as the default root filesystem format.

```
# Check filesystem health
sudo btrfs check /dev/sda9

# Scrub (background integrity check)
sudo btrfs scrub start /
sudo btrfs scrub status /

# Show disk usage
sudo btrfs filesystem df /
sudo btrfs filesystem usage /
```

## Manual Rollbacks

If an update causes problems, roll back to the inactive (previous) partition:

```
# Show partition priorities
cgpt show /dev/sda

# Prioritize the inactive USR partition (find which is inactive first)
cgpt prioritize -P 1 -i PARTITION_NUMBER /dev/sda

# Reboot into the previous version
sudo systemctl reboot
```

Alternatively, interrupt GRUB at boot and manually select the previous USR partition.

**Automatic rollback** occurs if the boot fails — GRUB tries the other partition automatically.

## Common Diagnostic Patterns

| Problem             | Command                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------|
| Unit won't start    | <code>journalctl -u SERVICE --no-pager</code>                                                      |
| Network issues      | <code>networkctl status</code> , <code>resolvectl status</code>                                    |
| Container issues    | <code>docker logs CONTAINER</code> , <code>journalctl CONTAINER_ID</code>                          |
| Boot issues         | <code>journalctl -b -1</code> (previous boot)                                                      |
| Ignition didn't run | Check <code>/var/log/ignition.log</code> ; verify <code>flatcar/first_boot</code> flag was present |
| Update stuck        | <code>update_engine_client -status</code> ; <code>journalctl -u update-engine</code>               |